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5

The OS and Process Management

Team Newton

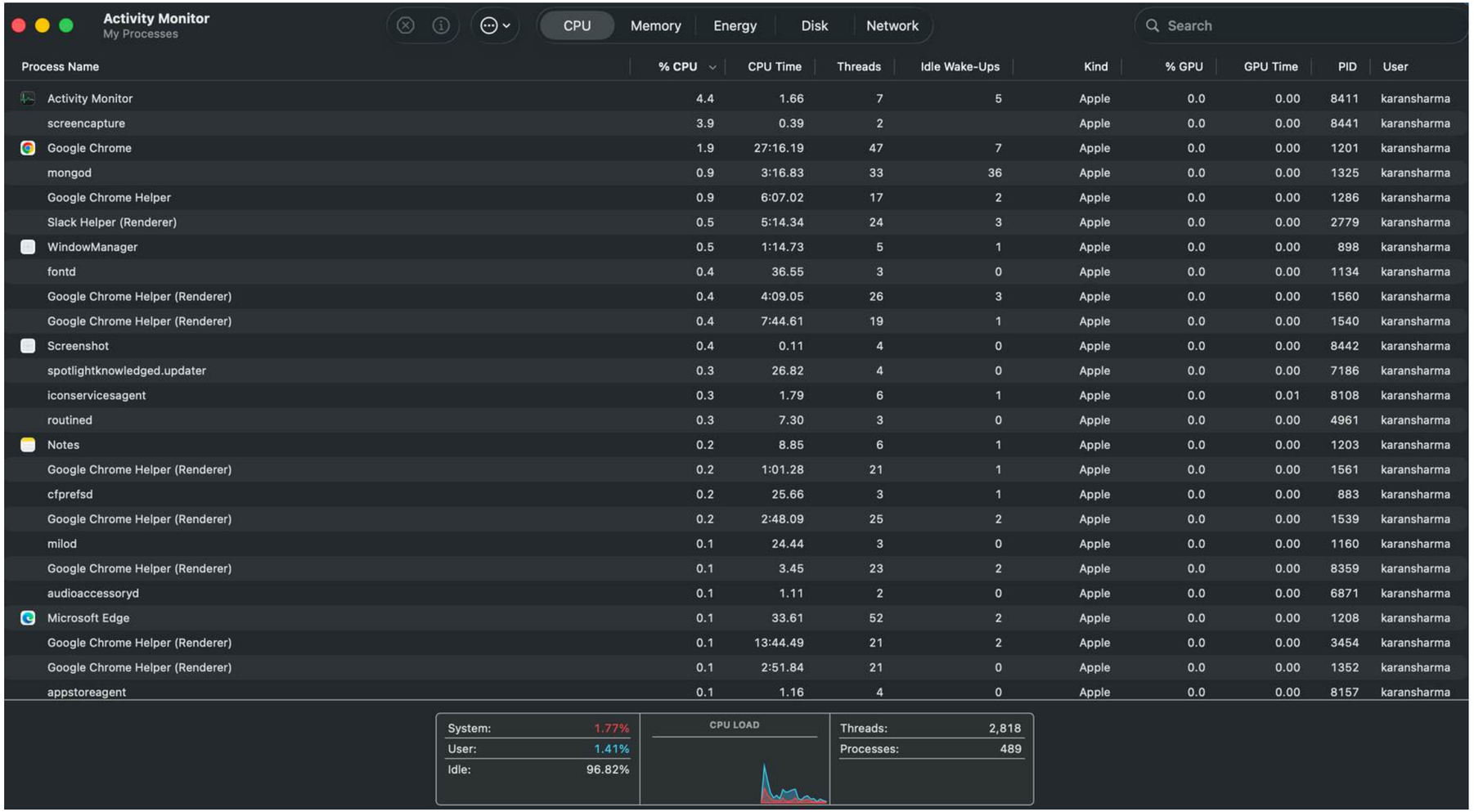
C05: System & AI Essentials

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Recap

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Process Name	% CPU	CPU Time	Threads	Idle Wake-Ups	Kind	% GPU	GPU Time	PID	User
Activity Monitor	4.4	1.66	7	5	Apple	0.0	0.00	8411	karansharma
screencapture	3.9	0.39	2		Apple	0.0	0.00	8441	karansharma
Google Chrome	1.9	27:16.19	47	7	Apple	0.0	0.00	1201	karansharma
mongod	0.9	3:16.83	33	36	Apple	0.0	0.00	1325	karansharma
Google Chrome Helper	0.9	6:07.02	17	2	Apple	0.0	0.00	1286	karansharma
Slack Helper (Renderer)	0.5	5:14.34	24	3	Apple	0.0	0.00	2779	karansharma
WindowManager	0.5	1:14.73	5	1	Apple	0.0	0.00	898	karansharma
fontd	0.4	36.55	3	0	Apple	0.0	0.00	1134	karansharma
Google Chrome Helper (Renderer)	0.4	4:09.05	26	3	Apple	0.0	0.00	1560	karansharma
Google Chrome Helper (Renderer)	0.4	7:44.61	19	1	Apple	0.0	0.00	1540	karansharma
Screenshot	0.4	0.11	4	0	Apple	0.0	0.00	8442	karansharma
spotlightknowledged.updater	0.3	26.82	4	0	Apple	0.0	0.00	7186	karansharma
iconservicesagent	0.3	1.79	6	1	Apple	0.0	0.01	8108	karansharma
routined	0.3	7.30	3	0	Apple	0.0	0.00	4961	karansharma
Notes	0.2	8.85	6	1	Apple	0.0	0.00	1203	karansharma
Google Chrome Helper (Renderer)	0.2	1:01.28	21	1	Apple	0.0	0.00	1561	karansharma
cfprefsd	0.2	25.66	3	1	Apple	0.0	0.00	883	karansharma
Google Chrome Helper (Renderer)	0.2	2:48.09	25	2	Apple	0.0	0.00	1539	karansharma
milod	0.1	24.44	3	0	Apple	0.0	0.00	1160	karansharma
Google Chrome Helper (Renderer)	0.1	3.45	23	2	Apple	0.0	0.00	8359	karansharma
audioaccessoryd	0.1	1.11	2	0	Apple	0.0	0.00	6871	karansharma
Microsoft Edge	0.1	33.61	52	2	Apple	0.0	0.00	1208	karansharma
Google Chrome Helper (Renderer)	0.1	13:44.49	21	2	Apple	0.0	0.00	3454	karansharma
Google Chrome Helper (Renderer)	0.1	2:51.84	21	0	Apple	0.0	0.00	1352	karansharma
appstoreagent	0.1	1.16	4	0	Apple	0.0	0.00	8157	karansharma

Your Computer Is Busy Everytime

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Your machine is doing dozens of things right now, and you did not ask it to do any of them. Something is in charge of all of this.

Your computer is currently:

- Running background services
- Managing memory
- Handling network traffic
- Monitoring hardware
- Scheduling CPU time

Who coordinates all of this?

The Operating System



What is an Operating System?

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An Operating System (OS) is a software that acts as a bridge between the **user** and the **computer hardware**. It manages all the software and hardware on a computer or digital device.

In simple words:
An operating system is like the **manager** of a computer. It tells the computer how to **run programs, manage files, and connect to devices** like keyboards, printers, or the internet.

The Operating System

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The Operating System

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Manage Process & Memory

The operating system creates, schedules, and manages running programs while allocating memory so multiple applications can run efficiently.

The Operating System

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Manage Process & Memory



Manage Devices

The operating system controls hardware devices such as the keyboard, mouse, printer, monitor, and storage through device drivers.

The Operating System

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Manage Process & Memory



Manage Devices



Manage Files & Storage

The operating system organizes files and folders, controls access to data, and manages storage devices like SSDs and hard drives.

The Operating System

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Manage Process & Memory



Manage Devices



Manage Files & Storage

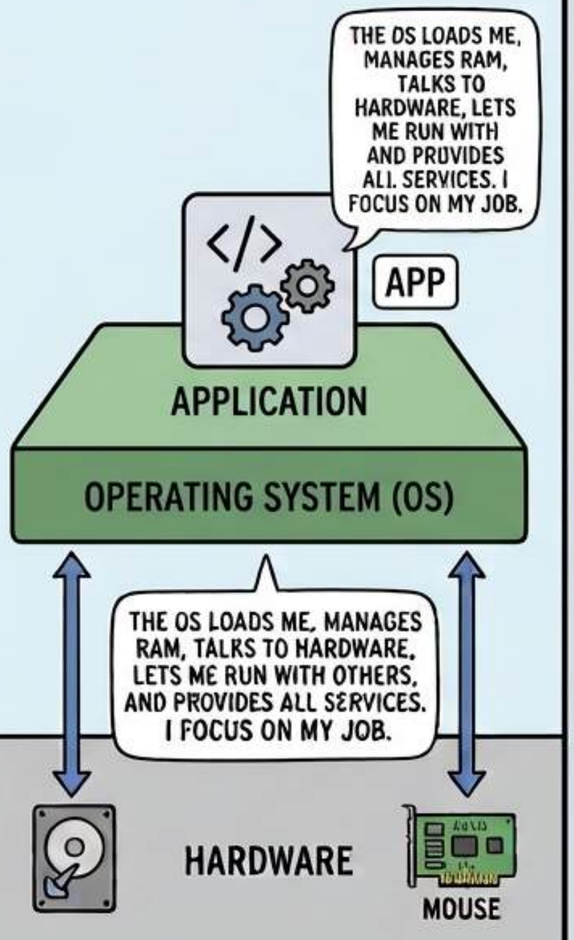


Provide User Interface

The operating system provides a graphical or command-line interface that allows users to interact with the computer and run applications.

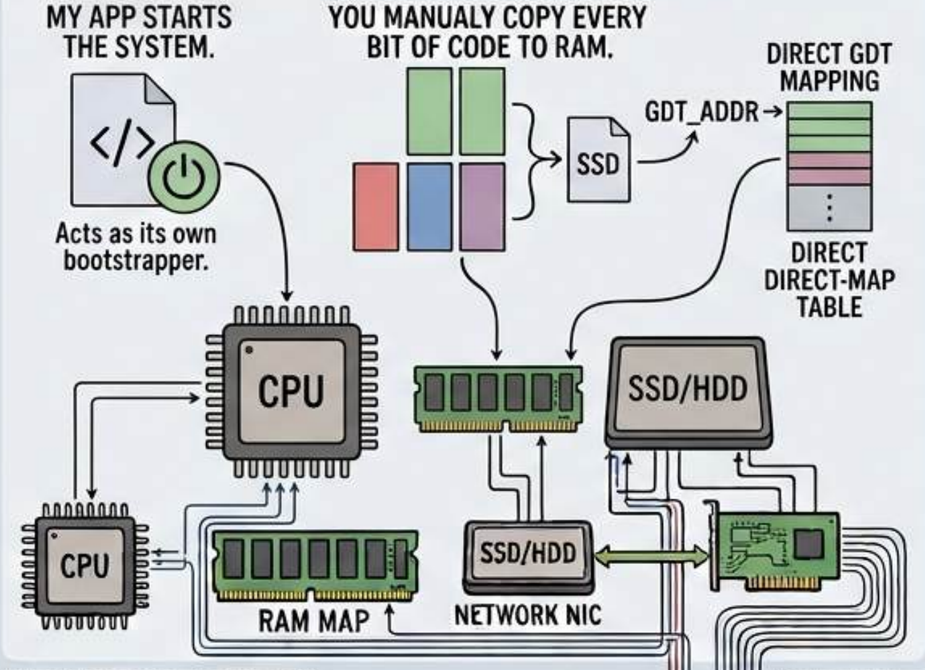
SOFTWARE: WITH OS vs. WITHOUT OS

WITH AN OS
(MODERN APP)

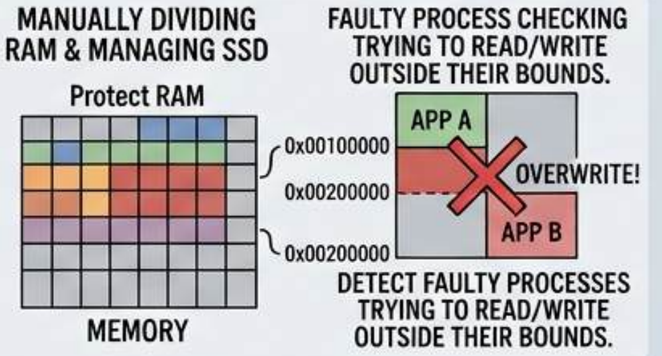


WITHOUT AN OS
(BARE METAL)

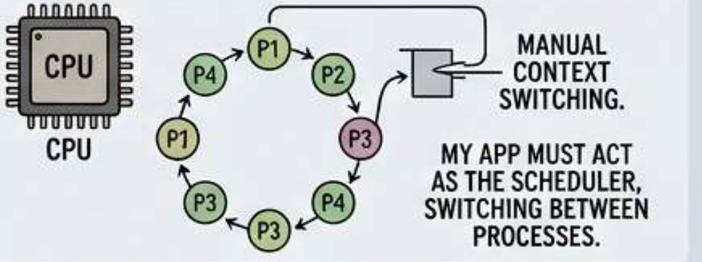
LOADING & STARTUP



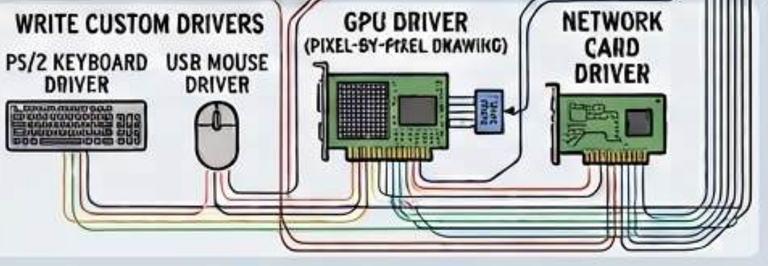
RAM & SSD MANAGEMENT



CPU PROCESS SCHEDULING



I/O HANDLING (DEVICES)



I/O HANDLING (BASIC)



TERMINATION & CLEANUP

MUST MANUALLY CLEAN UP.

- ☑ CLOSE FILES
- ☑ FREE ALL RAM
- ☑ RESET HARDWARE
- ☑ CLOSE CONNECTIONS

IF I FAIL, THE WHOLE SYSTEM LOCKS UP!

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Process Management

Recap of a program

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A program is a collection of **instructions** written in a programming language that tells a computer how to perform a specific task.

```
age = 20

if age >= 18:
    print("Eligible to vote")
else:
    print("Not eligible to vote")
```

Program

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One program on disk can create multiple **independent processes** in memory.

Example

- Program: Google Chrome (installed once)
- Action: Open two Chrome windows
- Result: The OS creates two separate Chrome processes, each with its own memory and resources.

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What happens in the background when you double click a word file?

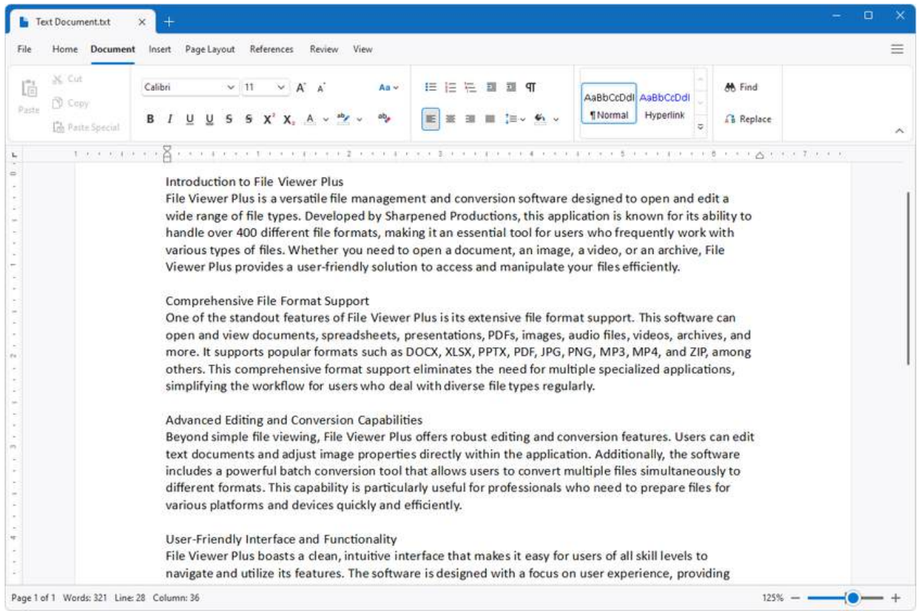
Process

A **process** is a program that is currently being executed by the computer.

When you run a program, the operating system loads it into **memory** and creates a process. A **process** contains the program's code, data, and the **resources** needed to execute it.



When you click on a icon like txt file, the program of the text editor comes under execution and hence you see the txt file.



Process

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- The Operating System (OS) finds the **program** on the disk.
- The OS loads the program into **memory** (RAM).
- The OS **creates** a process to run the program.
- The application **starts** and becomes usable.

One Process, Multiple Child Processes

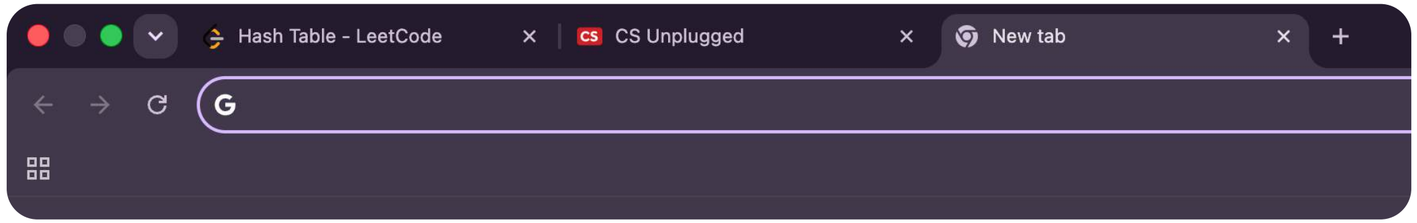
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Key Points

- **Main Process :** One main chrome process manages the browser.
- **Multiple Tabs as child Process :** Each tab run in its own process.
- **Independent execution :** If one tab crashes, other keep running smoothly.



Main Process

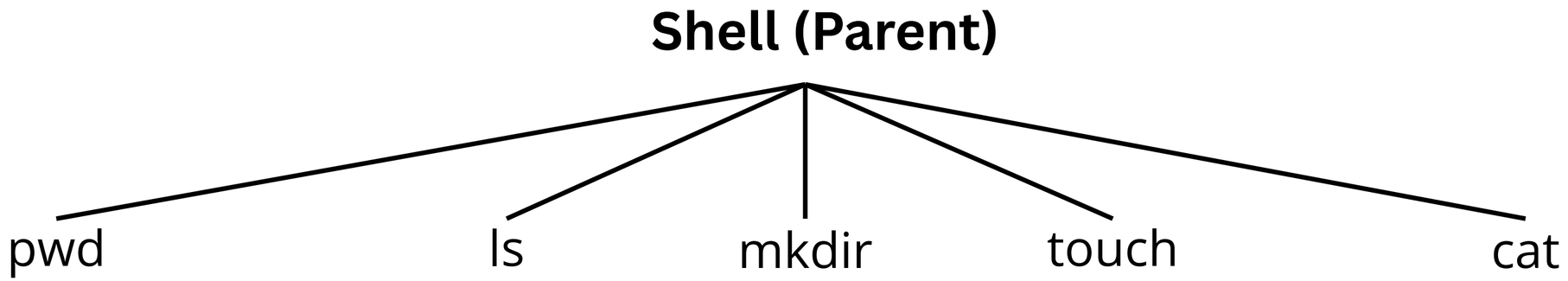


1 tab = 1 child process

Parent and Child Processes

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- **Parent Process:** A parent process is a process that creates one or more child processes using an operating system process creation mechanism.
- **Child Process:** A child process is a new process created by a parent process that executes independently while maintaining a relationship with its parent.



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How Can One CPU Run So Many Process?

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Timeslicing

Timeslicing

Time slicing is a scheduling technique where the CPU gives each process a small amount of CPU time, called a **time slice**.

The switch happens thousands of times every second.

To us, everything looks simultaneous.



Timeslicing

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| Chrome | VSCode | Spotify | Chrome | Slack | Python |



CPU

Time 0 ms → Chrome

Timeslicing

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| Chrome | VSCode | Spotify | Chrome | Slack | Python |



CPU

Time 0 ms → Chrome
Time 6 ms → VS Code

Timeslicing

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| Chrome | VSCode | Spotify | Chrome | Slack | Python |



CPU

Time 0 ms → Chrome
Time 6 ms → VS Code
Time 12 ms → Spotify

Timeslicing

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| Chrome | VSCode | Spotify | Chrome | Slack | Python |



CPU

Time 0 ms → Chrome
Time 6 ms → VS Code
Time 12 ms → Spotify
Time 18 ms → Chrome

Timeslicing

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| Chrome | VSCode | Spotify | Chrome | Slack | Python |



CPU

Time 0 ms → Chrome
Time 6 ms → VS Code
Time 12 ms → Spotify
Time 18 ms → Chrome
Time 24 ms → Slack

Timeslicing

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| Chrome | VSCode | Spotify | Chrome | Slack | Python |



CPU

Time 0 ms → Chrome
Time 6 ms → VS Code
Time 12 ms → Spotify
Time 18 ms → Chrome
Time 24 ms → Slack
Time 30 ms → Python

Timeslicing

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| Chrome | VSCode | Spotify | Chrome | Slack | Python |



CPU

Time 0 ms → Chrome
Time 6 ms → VS Code
Time 12 ms → Spotify
Time 18 ms → Chrome
Time 24 ms → Slack
Time 30 ms → Python

The CPU gives each process a tiny **time slice** before switching to the next one. This rapid switching creates the illusion that all applications are **running** simultaneously.

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When we opened Activity Monitor, we saw hundreds of processes.

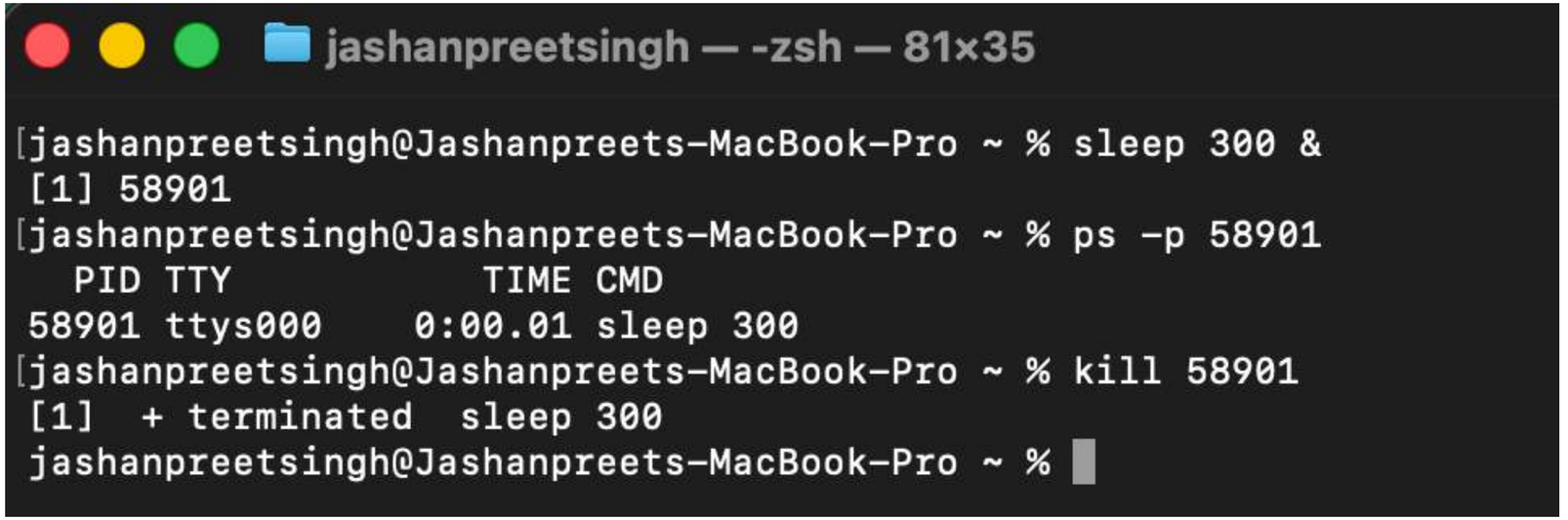
Can we see the same information without opening Activity Monitor?

kill: terminate a process

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The kill command is used to terminate a running process using its Process ID (PID).

- kill <PID> sends a SIGTERM (15) signal, politely requesting the process to stop and allowing it to clean up before exiting.
- kill -9 <PID> sends a SIGKILL (9) signal, forcing the process to terminate immediately without any cleanup.



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Activity

Creating a Runaway Process

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Goal

Observe how the OS handles a CPU-hungry program.



Steps

1. Run htop on terminal
2. Open Activity Monitor
3. Run Python script
4. Observe CPU usage
5. Kill the process

cpu-spike.py

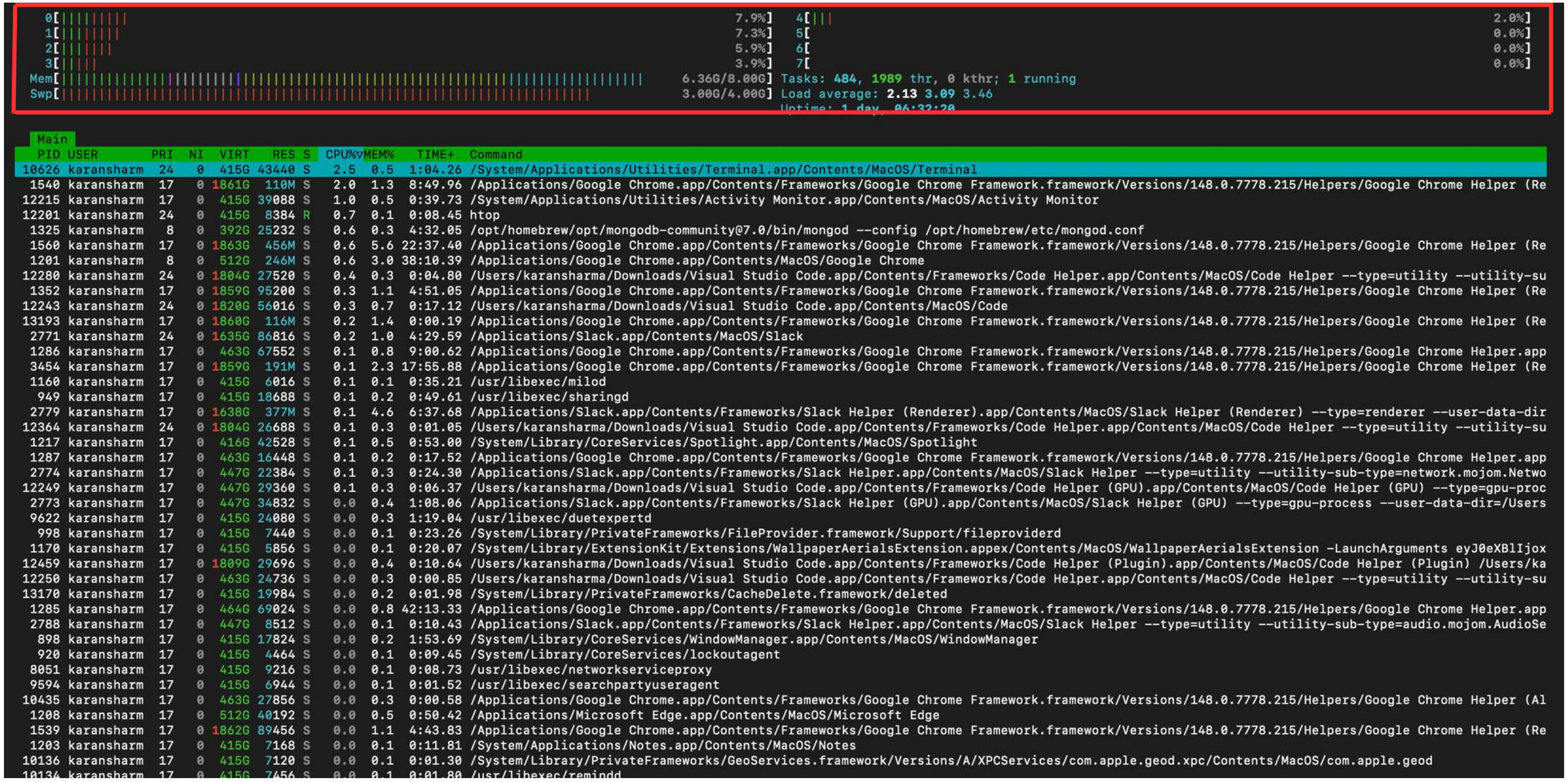
```python
import time
import multiprocessing

def burn(duration):
    end_time = time.time() + duration
    while time.time() < end_time:
        pass

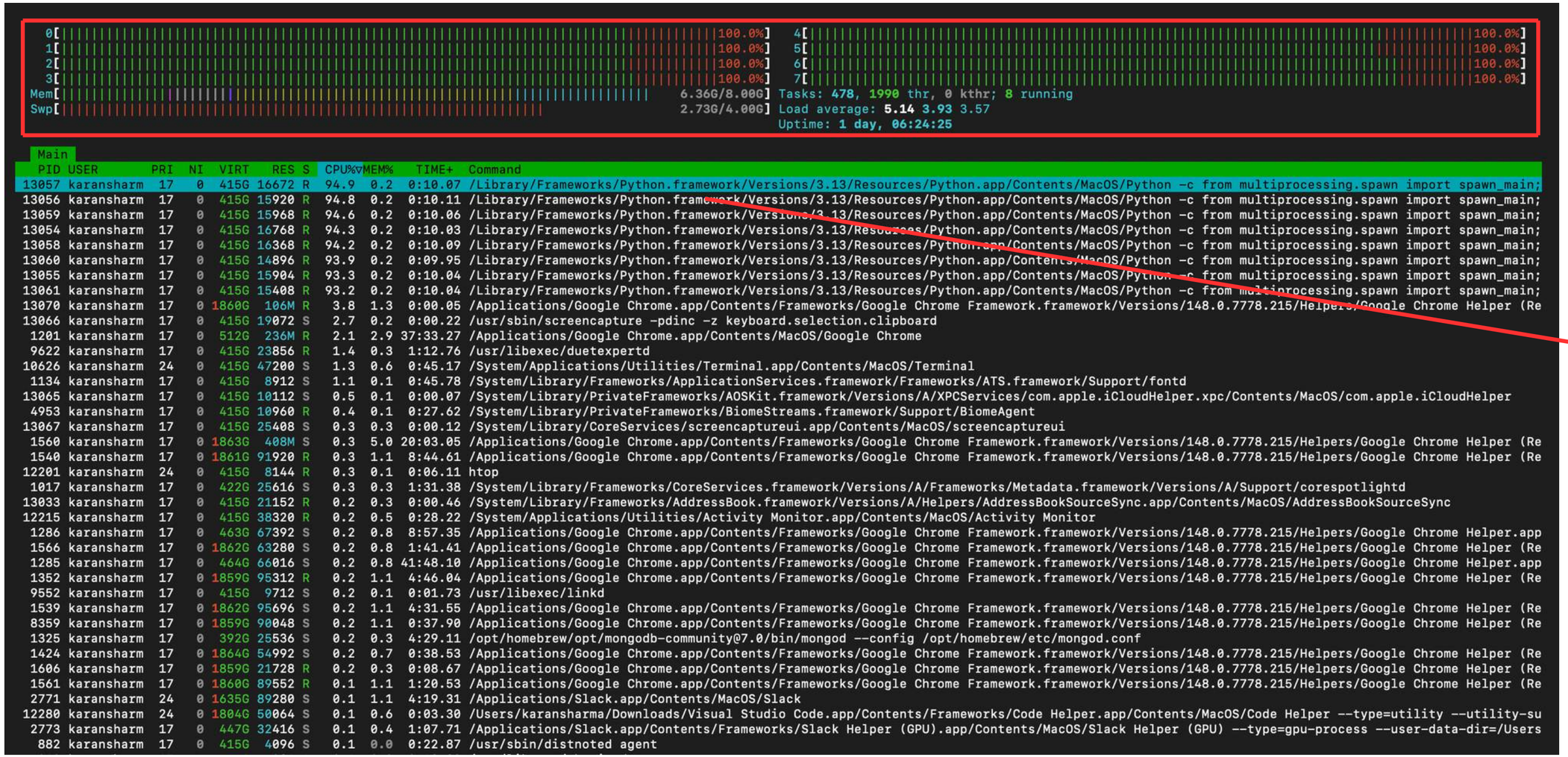
if __name__ == "__main__":
    duration = 120
    cores = multiprocessing.cpu_count()
    processes = [multiprocessing.Process(target=burn, args=(duration,)) for _ in range(cores)]
    for p in processes:
        p.start()
    for p in processes:
        p.join()
    print("done")

```

Before executing script

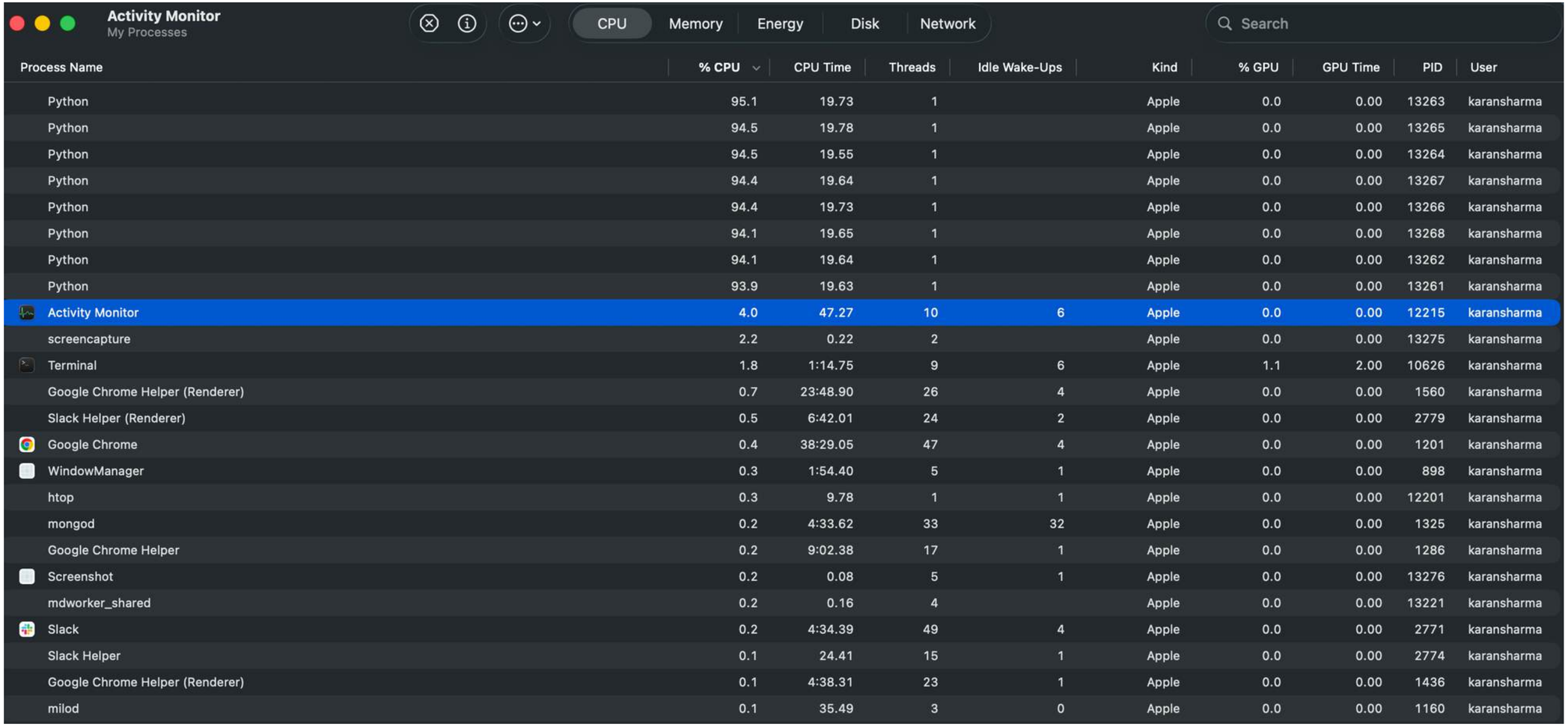

0[||||||||| 7.9%]
1[||||||||| 7.3%]
2[||||||| 5.9%]
3[||||| 3.9%]
4[||| 2.0%]
5[0.0%]
6[0.0%]
7[0.0%]
Mem[|||||||||

After Executing Script


0[|||||||

Activity Monitor

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Process Name	% CPU	CPU Time	Threads	Idle Wake-Ups	Kind	% GPU	GPU Time	PID	User
Python	95.1	19.73	1		Apple	0.0	0.00	13263	karansharma
Python	94.5	19.78	1		Apple	0.0	0.00	13265	karansharma
Python	94.5	19.55	1		Apple	0.0	0.00	13264	karansharma
Python	94.4	19.64	1		Apple	0.0	0.00	13267	karansharma
Python	94.4	19.73	1		Apple	0.0	0.00	13266	karansharma
Python	94.1	19.65	1		Apple	0.0	0.00	13268	karansharma
Python	94.1	19.64	1		Apple	0.0	0.00	13262	karansharma
Python	93.9	19.63	1		Apple	0.0	0.00	13261	karansharma
Activity Monitor	4.0	47.27	10	6	Apple	0.0	0.00	12215	karansharma
screencapture	2.2	0.22	2		Apple	0.0	0.00	13275	karansharma
Terminal	1.8	1:14.75	9	6	Apple	1.1	2.00	10626	karansharma
Google Chrome Helper (Renderer)	0.7	23:48.90	26	4	Apple	0.0	0.00	1560	karansharma
Slack Helper (Renderer)	0.5	6:42.01	24	2	Apple	0.0	0.00	2779	karansharma
Google Chrome	0.4	38:29.05	47	4	Apple	0.0	0.00	1201	karansharma
WindowManager	0.3	1:54.40	5	1	Apple	0.0	0.00	898	karansharma
htop	0.3	9.78	1	1	Apple	0.0	0.00	12201	karansharma
mongod	0.2	4:33.62	33	32	Apple	0.0	0.00	1325	karansharma
Google Chrome Helper	0.2	9:02.38	17	1	Apple	0.0	0.00	1286	karansharma
Screenshot	0.2	0.08	5	1	Apple	0.0	0.00	13276	karansharma
mdworker_shared	0.2	0.16	4		Apple	0.0	0.00	13221	karansharma
Slack	0.2	4:34.39	49	4	Apple	0.0	0.00	2771	karansharma
Slack Helper	0.1	24.41	15	1	Apple	0.0	0.00	2774	karansharma
Google Chrome Helper (Renderer)	0.1	4:38.31	23	1	Apple	0.0	0.00	1436	karansharma
milod	0.1	35.49	3	0	Apple	0.0	0.00	1160	karansharma

Result

Python processes move to the top. Highest CPU consumers.

Everything still runs

- Browser
- Terminal
- Spotify
- VS Code

Machine becomes slower.

But does not crash.

Killing the Process

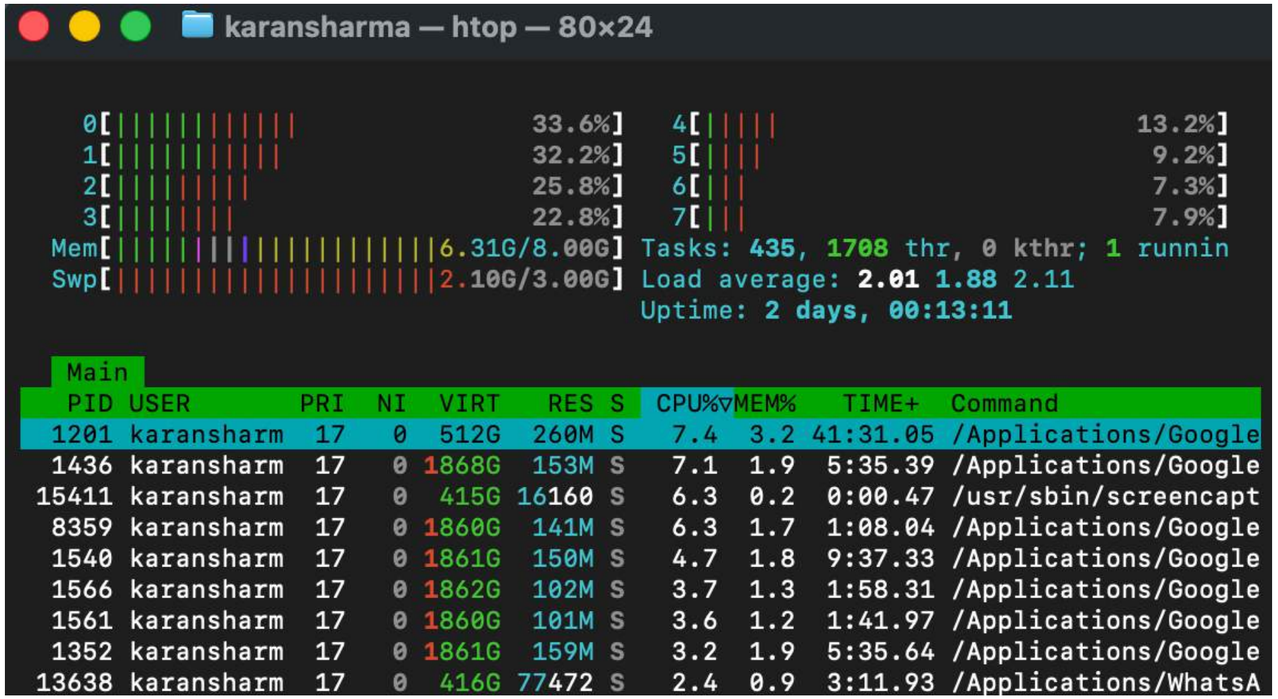
Find the PID from the htop

Command

kill -9 2341

Result

- Process disappears
- CPU drops immediately
- System returns to normal



Normal Again

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Memory Management

Every Process Gets Its Own Memory

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When the Operating System **creates** a process, it first **allocates** a portion of RAM to that process.

This memory is used to store:

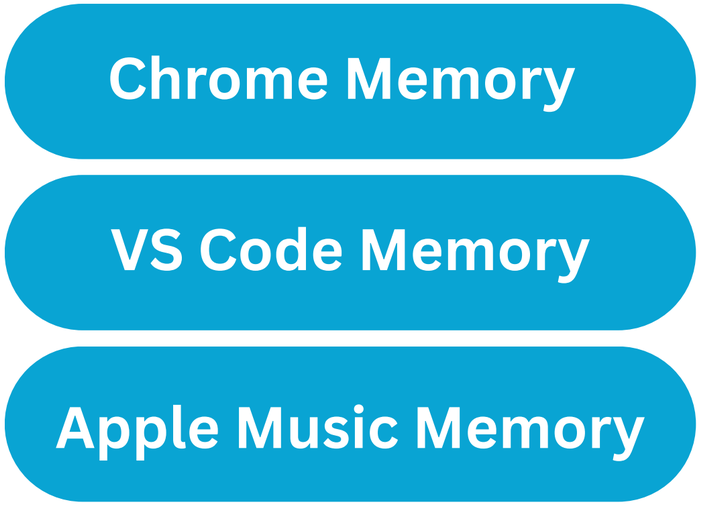
- The program's code
- Variables and data
- Function calls
- Temporary information needed while the program runs

The allocated memory belongs exclusively to that **process**.

Example

Suppose you have three applications running

The OS gives each application its own memory space.



Chrome cannot directly access VS Code's memory, and Apple Music cannot modify Chrome's data.

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Inclass Quiz

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Summary

Feedback

Now it's time to give your valuable feedback



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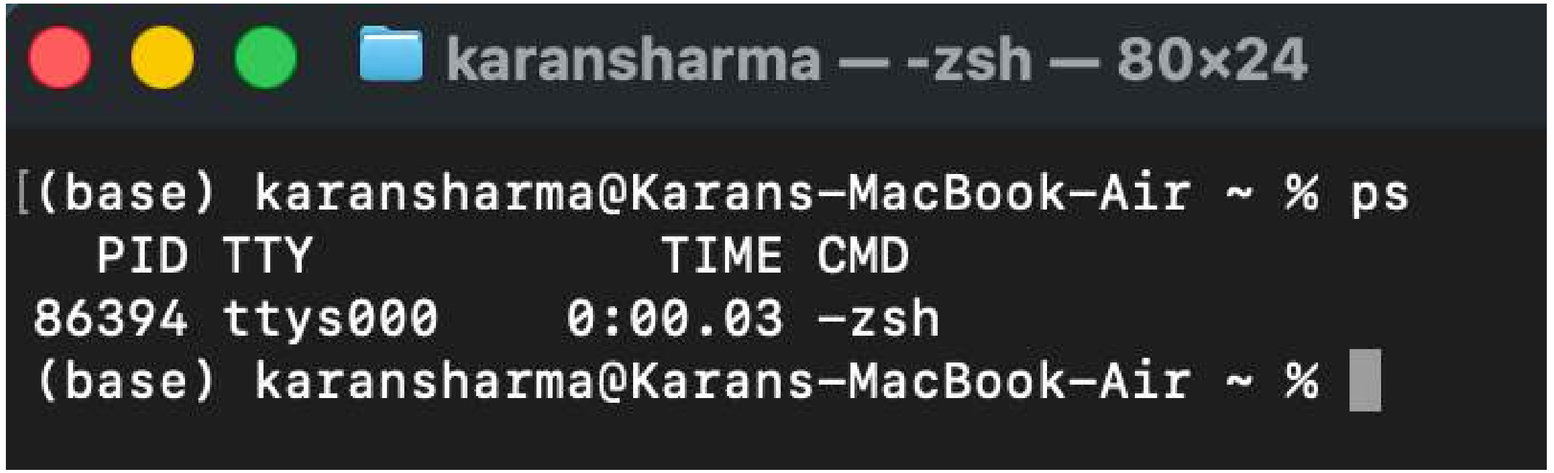
Thank You!

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Listing and Managing Processes using CLI

ps

ps stands for **Process Status**. It shows information about **running** processes.



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But there were so many processes we saw on the activity monitor. Where are they?

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But there were so many processes we saw on the activity monitor. Where are they?

ps only shows the processes **associated** with this Terminal session. It's showing only a small part of all the running processes.

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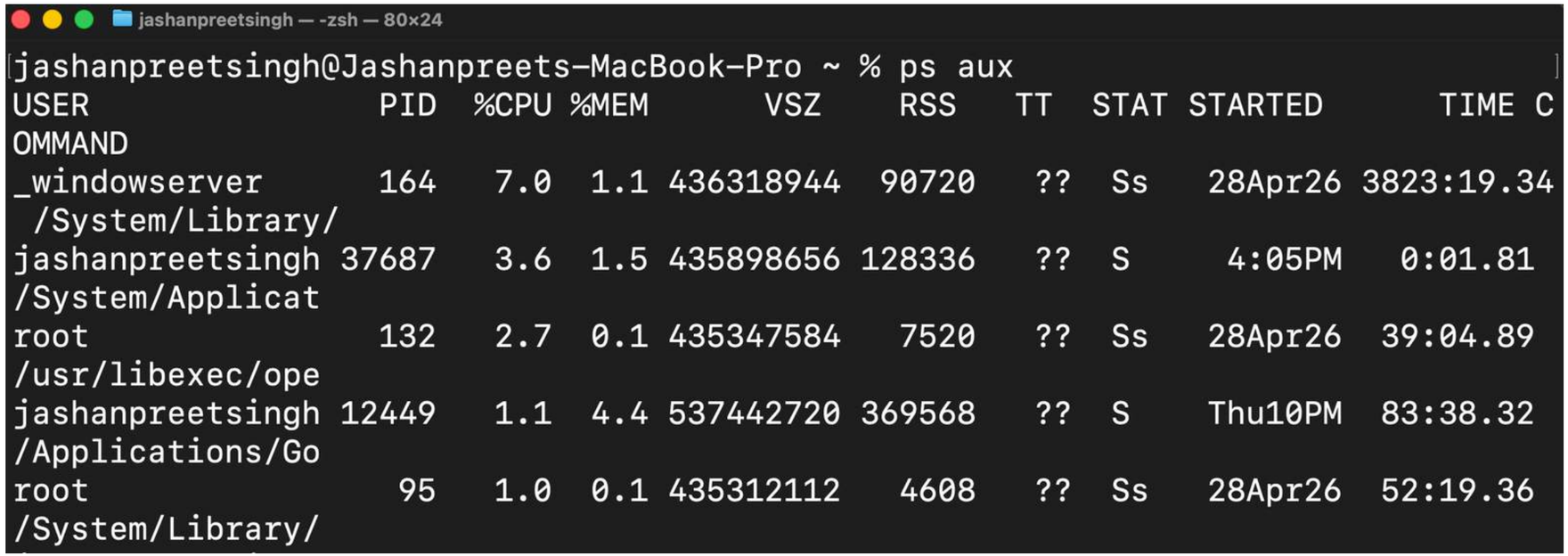
But there were so many processes we saw on the activity monitor. Where are they?

ps only shows the processes **associated** with this Terminal session. It's showing only a small part of all the running processes.

How do we see everything running on the computer?

ps aux

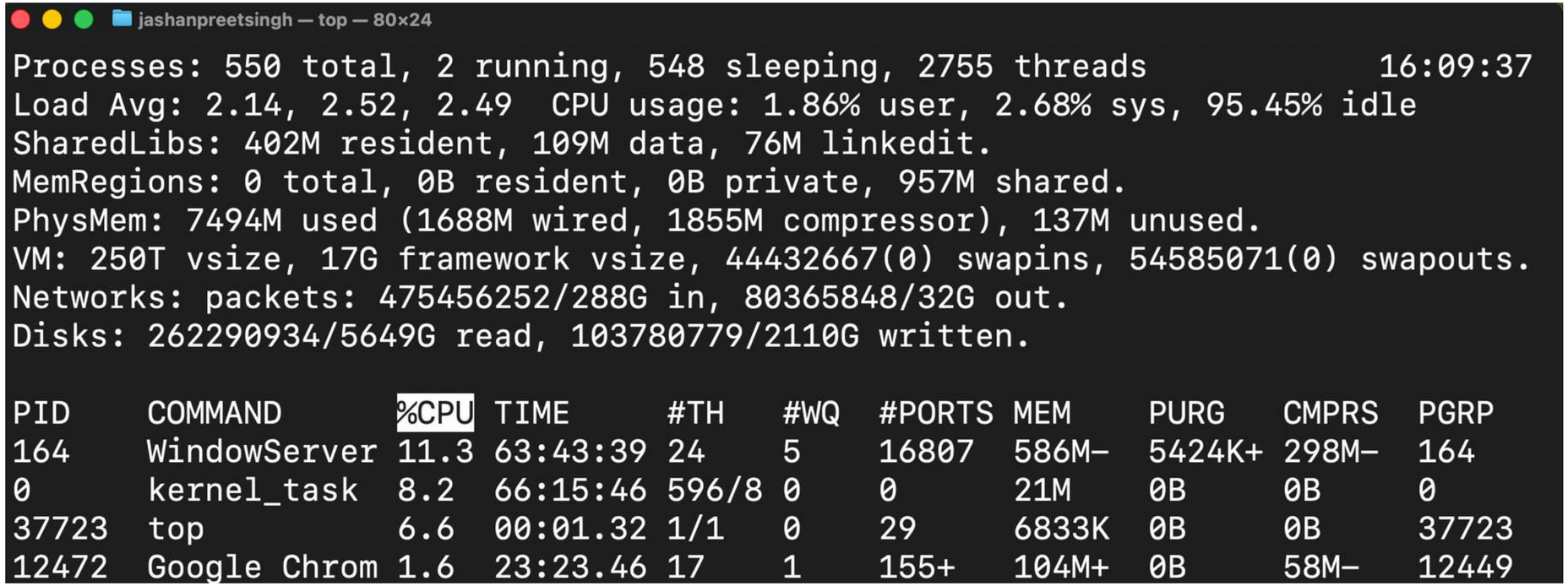
The **ps aux** command displays all processes running on the system. Its output includes several columns such as **USER** (the process owner), **PID** (Process ID), **%CPU** (CPU usage), **%MEM** (memory usage), and **COMMAND** (the program or command being executed).



top: live view

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The top command provides a live view of running processes, showing CPU and memory usage in real time. It continuously updates the information, making it useful for monitoring system activity. Press q to quit.



htop: a better view

The htop command provides a more interactive and user-friendly view of running processes. It displays CPU and memory usage with color-coded information and visual CPU bars, making system monitoring easier to understand. Press q to quit.

